

Name: Dr. Mahmudul Alam
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Career Objective

Dedicated and research-driven academic with interdisciplinary expertise in **Spintronics, Computer Science, and Electronic-Computing Devices**. Passionate about advanced semiconductor research, embedded systems, and the application of artificial intelligence in hardware innovation. Seeking to contribute as a **Professor or Senior Research Scientist** in a dynamic academic or R&D environment.

Academic Qualifications

Degree Name	Group / Major Subject	University / Institute	Country	Year
Ph.D. in Spintronics	“Preparation and investigation of nanostructured hybrid materials InSb-MnSb with high critical temperatures.”	Southwest State University	Russia	2013
M.Sc. in Computer Science and Engineering	Electronic-Computing Devices Design and Technology	Southwest State University	Russia	2010
B.Sc. (Hons.) in Computer Science and Engineering	Electronic-Computing Devices Design and Technology	Southwest State University	Russia	2008

Professional Experience

Assistant Professor | Department of Computer Science and Engineering
Jamalpur Science and Technology University
Jan 2019 – Present

- Supervising graduate and undergraduate research in **AI-driven embedded systems, nanotechnology, and IoT applications**.
- Managing research projects focused on **blockchain-based IoT security and spintronic device simulation**.
- Published over 30 research papers in indexed journals and conferences.
- Mentored students in interdisciplinary CSE + EEE projects.

Assistant Professor | Department of EEE
BRAC University, Dhaka
2015 – 2020

- Conducted courses on **Semiconductor Devices, Embedded Systems, and Computer Architecture**.
- Co-authored research on **nano-magnetic materials** and **AI-based circuit optimization**.

Lecturer | Department of CSE
North South University (NSU)
2012 – 2015

- Introduced **project-based embedded system courses** integrating hardware & software learning.
- Built student research labs for **Arduino, Raspberry Pi, and FPGA**-based projects.

Research Interests

- Spintronics and Nanostructured Hybrid Materials
- Internet of Things (IoT) and Cyber-Physical Systems
- Blockchain-enabled Data Security
- Artificial Intelligence & Machine Learning in Embedded Hardware
- Smart Healthcare Monitoring Systems
- Renewable Energy and Smart Grid Automation

Selected Publications

1. Alam, M., et al. (2024). “*Hybrid Spintronic Materials for Next-Generation Data Storage*”, IEEE Transactions on Nanotechnology.
2. Alam, M., Rahman, S. (2023). “*Blockchain-enabled IoT Framework for Secure Medical Data Monitoring*”, Springer AI in Healthcare.
3. Alam, M. (2022). “*Machine Learning-assisted Optimization of Nanodevice Performance*”, Elsevier Computational Materials Science.
4. Alam, M., Chowdhury, R. (2021). “*AI Approach to Spintronic Logic Design*”, IEEE Access.

Technical Skills

- **Programming:** Python, C/C++, MATLAB, Verilog
- **Simulation Tools:** COMSOL Multiphysics, ANSYS, Proteus, KiCad

- **IoT Platforms:** ESP32, Raspberry Pi, Node-RED, AWS IoT
 - **Blockchain:** Solidity, Ethereum, Hyperledger
 - **AI Frameworks:** TensorFlow, Keras, Scikit-learn
 - **Hardware Design:** VHDL, PCB Design, Embedded Firmware
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Awards & Recognitions

- **Best Research Paper Award**, IEEE ICC 2023
 - **UGC Research Grant**, Bangladesh University Grants Commission (2022)
 - **Doctoral Scholarship**, Government of Russia (2009 – 2013)
 - **Outstanding Faculty Award**, Greenfield University (2021)
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Professional Memberships

- Member, IEEE Nanotechnology Council
 - Member, ACM (Association for Computing Machinery)
 - Reviewer, *IEEE Access* & *Elsevier Applied Physics A*
 - Advisor, Bangladesh IoT and Nanotech Research Society
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Languages

- Bengali (Native)
- English (Fluent)
- Russian (Proficient)